

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B6. CONDENSED-MATTER PHYSICS

TRINITY TERM 2022

Thursday, 9th June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) The ‘auricupride’ form of the compound Cu_3Au has a primitive cubic lattice with $a = 0.375 \text{ nm}$. Au and Cu atoms are located, respectively, at the corners and the centre of the faces of the unit cell.

(i) Draw the unit cell and explain why this atomic arrangement is consistent with the stoichiometry of Cu_3Au .

(ii) Write the unit-cell structure factor of Cu_3Au and verify that it has the general form

$$S_{hkl} = f_{\text{Au}}(G_{hkl}) + \alpha_{hkl}f_{\text{Cu}}(G_{hkl})$$

where $f_X(G_{hkl})$ is the atomic form factor (X being any ion) and you must determine the numerical constant α_{hkl} for all possible hkl ’s.

(iii) In a scattering experiment performed with a wavelength λ , the hkl reflection is observed at a scattering angle $2\theta_{hkl}$. Show that G_{hkl} , λ and θ_{hkl} are related by the expression $G_{hkl} = 4\pi \sin \theta_{hkl} / \lambda$. [5]

(b) An X-ray powder experiment is performed on Cu_3Au with an X-ray wavelength $\lambda = 0.154 \text{ nm}$.

(i) Determine the scattering angle, 2θ , for the reflections (110) and (111).

(ii) Using the approximate expression $f_X(\sin \theta / \lambda) = Z_X \exp(-b_X(\sin \theta / \lambda)^2)$ with $Z_{\text{Cu}} = 29$, $Z_{\text{Au}} = 79$, $b_{\text{Cu}} = 6.3 \times 10^{-2} \text{ nm}^2$ and $b_{\text{Au}} = 8.0 \times 10^{-2} \text{ nm}^2$, determine the ratio of the intensities I_{111}/I_{110} , including the multiplicity but disregarding any geometrical constant of the diffractometer as well as the Lorentz polarisation factor. [7]

(c) Cu_3Au can also crystallise as a completely disordered alloy with approximately the same conventional unit cell and atoms at the same sites, each randomly populated by Au and Cu. In this case, the atomic form factor is identical for all sites and is the statistical average of f_{Au} and f_{Cu} . (i) Show that in this case $I_{110} = 0$ while I_{111} is the same as for the auricupride form. (ii) Explain why this is consistent with the extinction rules of this ‘average’ crystal structure [*Hint: it may be useful to modify your previous drawing of the crystal structure*]. [6]

(d) By suitable heat treatments, it is possible to crystallise Cu_3Au with different degrees of ordering, from fully-ordered auricupride to the fully disordered alloy. Introducing a parameter ρ that describes the degree of ordering (fully ordered: $\rho = 1$, fully disordered: $\rho = 0$),

(i) show that the average form factor of the atomic site at the corners of the unit cell can be written as $\langle f_{\text{corners}} \rangle = \frac{1}{4} [(3\rho + 1)f_{\text{Au}} + 3(1 - \rho)f_{\text{Cu}}]$ and derive the average form factor at the unit cell faces $\langle f_{\text{faces}} \rangle$.

(ii) Show that I_{111} is independent of ρ while I_{110} is proportional to ρ^2 . [7]

2. In the nearly-free-electron model, the electronic dispersion in one dimension for the first two bands can be well approximated near the zone boundary by the formula:

$$E = E_{\text{ZB}} \left(1 + y^2 \pm \sqrt{4y^2 + \epsilon^2} \right)$$

where $y = 1 - ka/\pi$ ($0 \leq y \leq 2$), k is the wavenumber and $\epsilon = |V_G|/E_{\text{ZB}}$.

(a) (i) Define the quantity E_{ZB} in terms of fundamental constants and the lattice parameter a . (ii) Define the relation between V_G and the electrostatic potential and determine the energy gap at the zone boundary. (iii) Draw a sketch of the dispersion for the cases $V_G = 0$ and $V_G \neq 0$. [5]

(b) (i) Calculate the total electronic energy density (energy per unit length) for a completely filled lower band (2 electrons per unit cell), assuming that, for this purpose, the dispersion in part (a) can be taken as valid in the interval $-1 \leq y \leq 1$. [The following integral is useful: $\int \sqrt{y^2 + c^2} dy = \frac{1}{2}y\sqrt{c^2 + y^2} + \frac{1}{2}c^2 \log \left(y + \sqrt{c^2 + y^2} \right)$.]

(ii) Show that this energy density is always less than for the case $\epsilon = 0$, and that this is consistent with your sketch of the dispersion.

(iii) Show that in the limit $\epsilon \ll 1$, the energy density has the following expression:

$$\frac{E_{\text{ZB}}}{a} \left(\frac{2}{3} - \frac{\epsilon^2}{4} \left(\alpha - \beta \log \frac{\epsilon}{4} \right) \right)$$

where α and β are numerical constants to be determined. [8]

(c) Consider a one-dimensional (1D) chain material with two atoms per unit cell, each contributing one electron. The atoms are at fractional positions $1/4 - \delta$ and $3/4 + \delta$ where $\delta \ll 1$, so that there is an alternation of shorter and longer distances between atoms (in this situation, the chain is said to be ‘dimerised’). A simple model for such a material can be obtained with the following potential, which has minima at the stated atomic positions:

$$V(x) = V_0 \left[\cos \left(\frac{4\pi}{a} x \right) - 2\pi\delta \left(\cos \left(\frac{2\pi}{a} x \right) - \cos \left(\frac{6\pi}{a} x \right) \right) \right].$$

(i) Determine V_G for such potential, and show that the energy gap is proportional to $|\delta|$ and is therefore zero for $\delta = 0$. (ii) Explain why this is consistent with the absence of a gap in a non-dimerised chain with one electron per unit cell. (iii) Using the results of part (b), write the expression for the electronic energy density as a function of δ . [7]

(d) For a realistic atomic chain, displacing the atoms from $\delta = 0$ (non-dimerised chain) to $\delta \neq 0$ (dimerised chain) incurs an elastic energy cost $E_{\text{elastic}} = \kappa\delta^2$, where κ is a positive elastic constant. Without doing any calculation, show that the equilibrium value of δ that minimises the total energy (elastic + electronic) is always > 0 , implying that such a chain would always dimerise spontaneously. [5]

3. (a) The number densities of thermally excited electrons and holes in a semiconductor are given by the formulas:

$$n(T) = \int_{\epsilon_c}^{\infty} g_c(\epsilon) f_c(\epsilon) d\epsilon ; \quad p(T) = \int_{-\infty}^{\epsilon_v} g_v(\epsilon) f_v(\epsilon) d\epsilon$$

where

$$g_c(\epsilon) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3} (\epsilon - \epsilon_c)^{1/2} ; \quad g_v(\epsilon) = \frac{(2m_h)^{3/2}}{2\pi^2\hbar^3} (\epsilon_v - \epsilon)^{1/2}$$

are densities of states in the conduction and valence bands, m_e , m_h , ϵ_v and ϵ_c have the usual meaning, $f_c(\epsilon)$, $f_v(\epsilon)$ are appropriate statistical factors and $E_g = \epsilon_c - \epsilon_v$. Use these expressions to derive the law of mass action, defining all quantities and stating any approximation you are making. [The following integral is useful: $\int_0^{\infty} \xi^{1/2} e^{-\xi} d\xi = \frac{\sqrt{\pi}}{2}$.]

[7]

(b) Consider a semiconductor doped with a single donor species with concentration N_D , each donor contributing a single electron to the conduction band when ionised. The donors create an atomic-like state below the conduction band, at an energy ϵ_D .

(i) By employing Fermi statistics, show that the ratio between occupied and unoccupied impurity levels is given by the following formula (μ is the chemical potential):

$$\frac{n_{\text{full}}}{n_{\text{empty}}} = e^{\beta(\mu - \epsilon_D)}.$$

(ii) Using the results from part (a) for the conduction electrons and from part (b)(i) for the impurity states, show that at low temperatures ($\beta^{-1} = k_b T \ll E_g$) the electron concentration n in the conduction band is given by:

$$\frac{n^2}{N_D - n} = 2 \left(\frac{m_e}{2\pi\beta\hbar^2} \right)^{\frac{3}{2}} e^{-\beta E_i}$$

where $E_i = \epsilon_c - \epsilon_D$ is the binding energy of the impurity states.

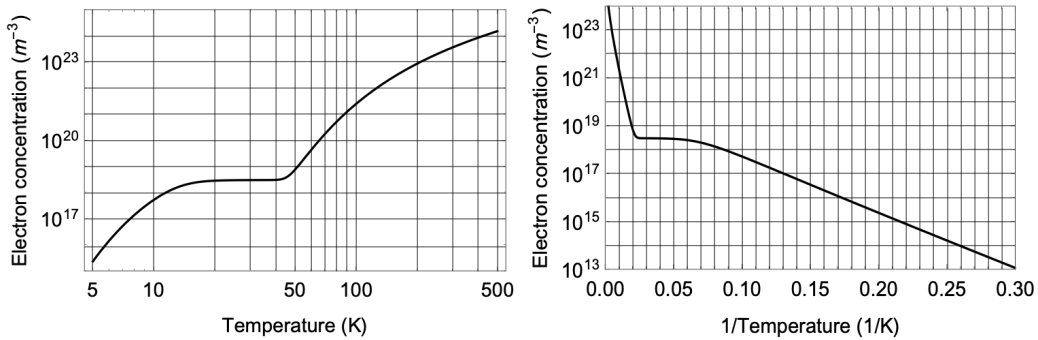
[10]

(c) The figures below display the electron density n in the conduction band of a doped semiconductor as in part (b), plotted on two different scales.

(i) Explain the features in the curves.

(ii) Use the figures to deduce the donor concentration N_D , the binding energy E_i and the band gap E_g .

[8]



4. (a) State the Hund's rules for a free ion and explain their origin in terms of microscopic interactions and couplings. [3]

(b) The molar susceptibility in the paramagnetic state of a compound with a single magnetic ion per unit formula can be expressed as:

$$\chi_{mol} = \left(1.57 \times 10^{-6} \text{ m}^3 \text{ K mol}^{-1}\right) \frac{(m_{\text{eff}})^2}{T - \Theta}$$

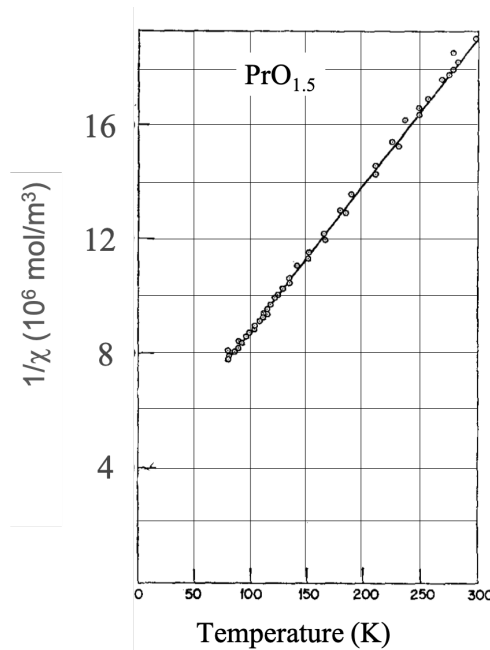
where m_{eff} is the effective magnetic moment in units of the Bohr magneton μ_B and T is the temperature.

(i) Derive this expression from the general expression for the susceptibility per unit volume (which you may quote).

(ii) Write the relation between m_{eff} , the Landé factor $\tilde{g}$ and the total angular momentum J .

(iii) Define the quantity Θ and state its relation to the sign of the interaction between magnetic moments. [7]

(c) The figure below displays the inverse of the molar magnetic susceptibility of a compound with chemical formula $\text{PrO}_{1.5}$ in which Pr is in the 3+ oxidation state.



From these data, extract m_{eff} for Pr^{3+} and Θ , stating whether the interactions between Pr^{3+} ions are ferromagnetic or antiferromagnetic as per part (b). [8]

(d) (i) Using Hund's rules and the Aufbau principle, determine L , S , and J for an isolated ion of Pr^{3+} (electronic configuration: $[\text{Xe}]4f^2$).

(ii) Using the value of m_{eff} you determined in part (c), calculate the Landé factor $\tilde{g}$ and compare it with the theoretical value for an isolated ion:

$$\tilde{g} = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}.$$